

```

1  #include <stdio.h>
2  # PRIORITY WITH AT #####
3  int main()
4  {
5      int n;
6      printf("Enter number of processes: ");
7      scanf("%d", &n);
8
9      int arrivalTime[10], burstTime[10], priority[10];
10     int remainingTime[10], waitingTime[10] = {0}, turnaroundTime[10];
11     int completed = 0, currentTime = 0;
12
13     for(int i = 0; i < n; i++)
14     {
15         printf("Enter AT, BT, PR for P%d: ", i+1);
16         scanf("%d%d%d", &arrivalTime[i],
17                 &burstTime[i],
18                 &priority[i]);
19
20         remainingTime[i] = burstTime[i];
21     }
22
23     while(completed < n)
24     {
25         int selected = -1;
26         int highestPriority = 9999;
27
28         for(int i = 0; i < n; i++)
29         {
30             if(arrivalTime[i] <= currentTime &&
31                 remainingTime[i] > 0 &&
32                 priority[i] < highestPriority)
33             {
34                 highestPriority = priority[i];
35                 selected = i;
36             }
37         }
38
39         if(selected == -1)
40         {
41             currentTime++; // CPU idle
42             continue;
43         }
44
45         remainingTime[selected]--;
46         currentTime++;
47
48         if(remainingTime[selected] == 0)
49         {
50             completed++;
51             turnaroundTime[selected] =
52                 currentTime - arrivalTime[selected];
53
54             waitingTime[selected] =
55                 turnaroundTime[selected] - burstTime[selected];
56         }
57     }
58
59     int totalWT = 0, totalTAT = 0;
60
61     printf("\nPID\tAT\tBT\tPR\tWT\tTAT\n");
62     for(int i = 0; i < n; i++)
63     {
64         printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
65                 i+1, arrivalTime[i], burstTime[i],
66                 priority[i], waitingTime[i], turnaroundTime[i]);
67
68         totalWT += waitingTime[i];
69         totalTAT += turnaroundTime[i];
70     }
71
72     printf("Average WT = %.2f\n", (float)totalWT / n);
73     printf("Average TAT = %.2f\n", (float)totalTAT / n);
74
75 }

```